

Guest Identity, Reservation & Digital Key Validator Smart Contract

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Overview

This unified Plutus V2 smart contract manages a guest's complete hotel stay life cycle, including identity submission, room reservation, digital key generation and validation, and final check-out. All actions are authorized via admin-signed transactions and validated using nested datum structures to reduce memory footprint and ensure precise validation at each stage.

Data Structures

Datum: GuestDatum

Represents the complete state of a guest's interaction with the system.

Field	Type	Description
guestAddress	BuiltinByteString	Guest's unique ID/address
adminPKH	PubKeyHash	Admin wallet that must sign transactions

identity	IdentityInfo	Identity fields nested in structure
reservation	ReservationInfo	Reservation details
keyInfo	KeyInfo	Digital Key , check-in-state , and flags

IdentityInfo

Field	Type	Description
name	BuiltinByteString	Guest name
passportNumber	BuiltinByteString	Passport number
photoHash	BuiltinByteString	Hash of identity photo
isUserVerified	Bool	Set true by admin
identityStatus	Bool	True if identity setup complete

ReservationInfo

Field	Type	Description
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isReserved	Bool	Whether reservation is initiated
reservationStatus	Bool	Whether reservation is confirmed
reservationId	BuiltinByteString	Unique reservation identifier
checkInDate	BuiltinByteString	Guest's check-in-date
checkOutDate	BuiltinByteString	Guest's check-out-date

KeyInfo

Field	Type	Description
initiateCheckIn	Bool	True if guest is ready to check in
DigitalKey	BuiltinByteString	Digital key issued to the guest
isDigitalKeyValidated	Bool	Validation Status of the digital key

Redeemer: GuestRedeemer

Used to determine which part of the logic the validator should enforce.

- FullReservation
- FullIdentitySubmit
- GenerateAndValidateKey
- CheckOut

Validation Logic by Redeemer

1.FullReservation

Actor: Admin only

Purpose: Sets room reservation data for the guest.

Preconditions:

- Guest address must be non-empty
- No existing reservation (isReserved == False)

Expected Changes:

- isReserved = True
- reservationStatus = True
- All reservation fields populated
- Other fields (identity, keyInfo, adminPKH) remain unchanged

2. FullIdentitySubmit

Actor: Admin only

Purpose: Captures guest identity data and marks them verified.

Preconditions:

- Guest must already have a reservation (isReserved == True)
- Existing identity fields must be empty

Expected Changes:

- Fields name, passportNumber, photoHash filled
- isUserVerified = True, identityStatus = True
- Other fields remain unchanged

3. GenerateAndValidateKey

Actor: Admin only

Purpose: Generates and validates the digital room key.

Preconditions:

- Guest has an active reservation (isReserved == True)
- Identity must be verified (isUserVerified == True)
- No digital key exists (digitalKey == "")

Expected Changes:

- digitalKey is populated
- initiateCheckIn = True
- isDigitalKeyValidated = True

- Other fields remain unchanged

4. CheckOut

Actor: Admin only

Purpose: Final checkout process to clear reservation and key info.

Preconditions:

- Guest must have been checked in (key and reservation were active)

Expected Changes:

- All reservation fields reset to default/empty
- digitalKey = "", isDigitalKeyValidated = False, initiateCheckIn = False
- Identity and admin fields remain unchanged

Smart Contract Low-Level Diagram

https://drive.google.com/drive/folders/1-ZhYD2nWvAH_k7zIyTv42eteB8uhiu59

*please download and open it in the browser

